5 Conclusions That Would Kill Your Policy

A Data Integrity Analysis of Global Health Statistics

Team importNumpy

DND Spring 2026 Datathon · Cal Poly Pomona

Dataset: Global Health Statistics · 1,000,000 rows · 20 countries · 2000–2024

What's in the Data — and What's Not

MEASURED ✓

- Outcomes: mortality_rate, recovery_rate, DALYs
- Access proxies: healthcare_access, doctors_per_1000, hospital_beds_per_1000, resource_index
- Socioeconomic: per_capita_income, education_index
- Geography proxy: urbanization_rate → 3 tier schemes
- Structure: country, year, disease, age_group, gender
- Scale: 1M rows · 119,978 unique combinations

NOT MEASURED ✗

- Distance-to-care (travel time, proximity)
- Facility locations or density maps
- Utilization (visits, admissions, claims)
- Continuity-of-care metrics
- Real-world reporting variation or missingness

All claims are labeled: DESCRIPTIVE | ROBUSTNESS | ILLUSTRATIVE — see appendix for evidence contract.

DESCRIPTIVE

Clean Data ≠ Realistic Data — Six Synthetic Signatures

DATA QUALITY ✓

- 0% missingness across all 25 columns
- 0 out-of-range values on all rate columns
- 1,000,000 rows, all pass quality flags
- 119,978 / 120,000 possible combinations present (99.98%)
- All 20 disease names correctly mapped

SYNTHETIC SIGNATURES ⚠

⚠ Near-Cartesian coverage: 99.98% of all (country × year × disease × age × gender) cells filled

⚠ Near-zero dependence: |r|max = 0.0023 across 14 variables

⚠ Bounded-range match: observed mean/SD match Uniform(a,b) expectations to <0.01% error

⚠ Homogeneous countries: mortality SD across 20 countries = 0.013 (ANOVA η^2 = 0.0000208)

⚠ Randomized disease labels: entropy = 3.459 ≈ theoretical max

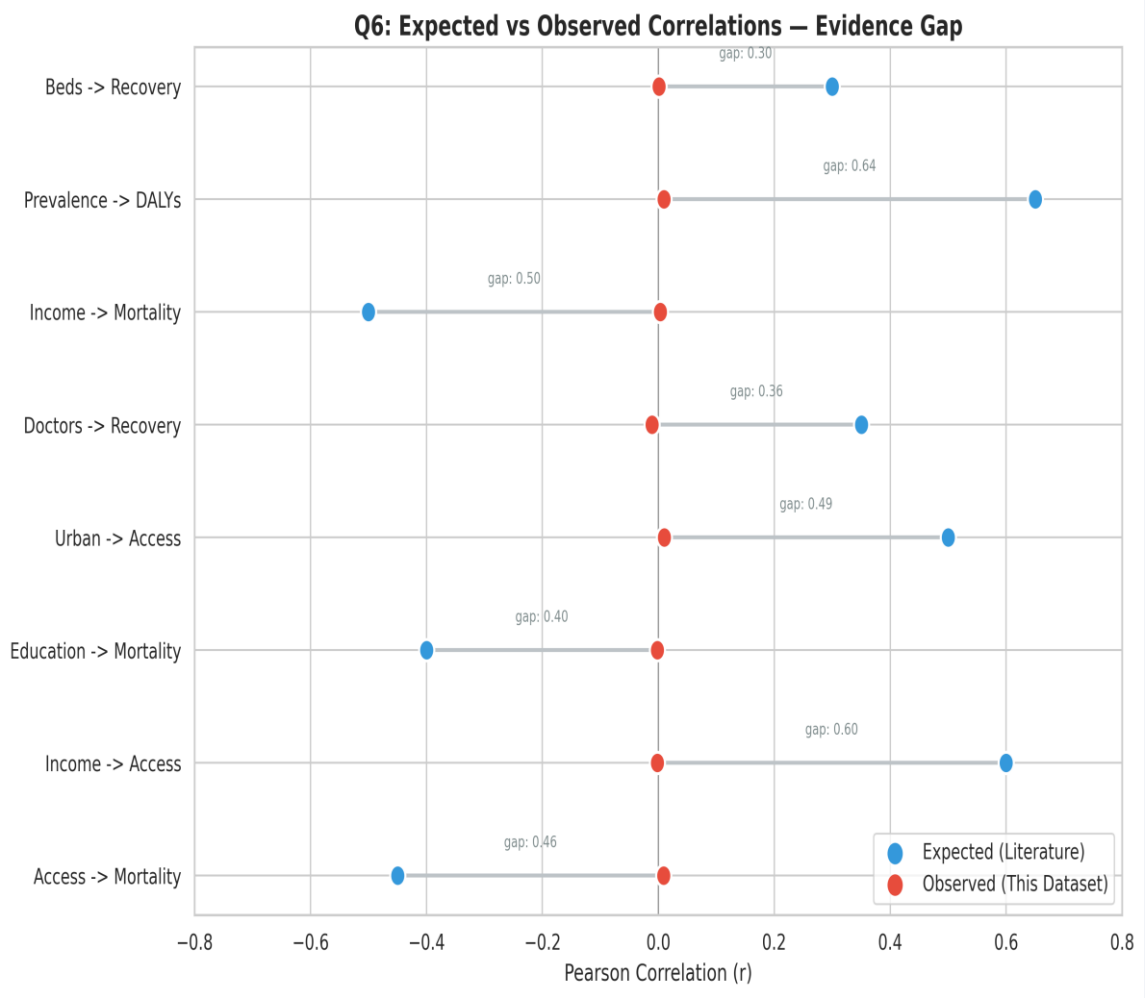
⚠ Strict 2-decimal quantization: 100% of numeric values

DESCRIPTIVE

Geography Explains 0.002% of Mortality Variance

Metric	This Dataset	Real-World Expected
Access ↔ Mortality (r)	0.00008	−0.30 to −0.50
Income ↔ Mortality (r)	−0.0015	−0.20 to −0.40
Country ANOVA η^2	0.0000208	0.05 to 0.30
Year trend η^2	0.0000106	0.01 to 0.15

Expected ranges from WHO/Lancet global health literature. Observed values are descriptive of this dataset only.



DESCRIPTIVE

ILLUSTRATIVE

“Rural Areas Have Worse Outcomes” — Evidence: ABSENT

Common claim:

Urban/rural classification predicts health outcomes.

What we found (DESCRIPTIVE):

- Cohen’s d < 0.01 for all tier pairs (negligible effect)
- ANOVA η^2 = 0.0000208 (geography explains 0.002%)
- p < 0.001 — but N=1M makes everything significant
- Absolute mortality difference Rural vs Urban: < 0.05pp

Why it’s premature:

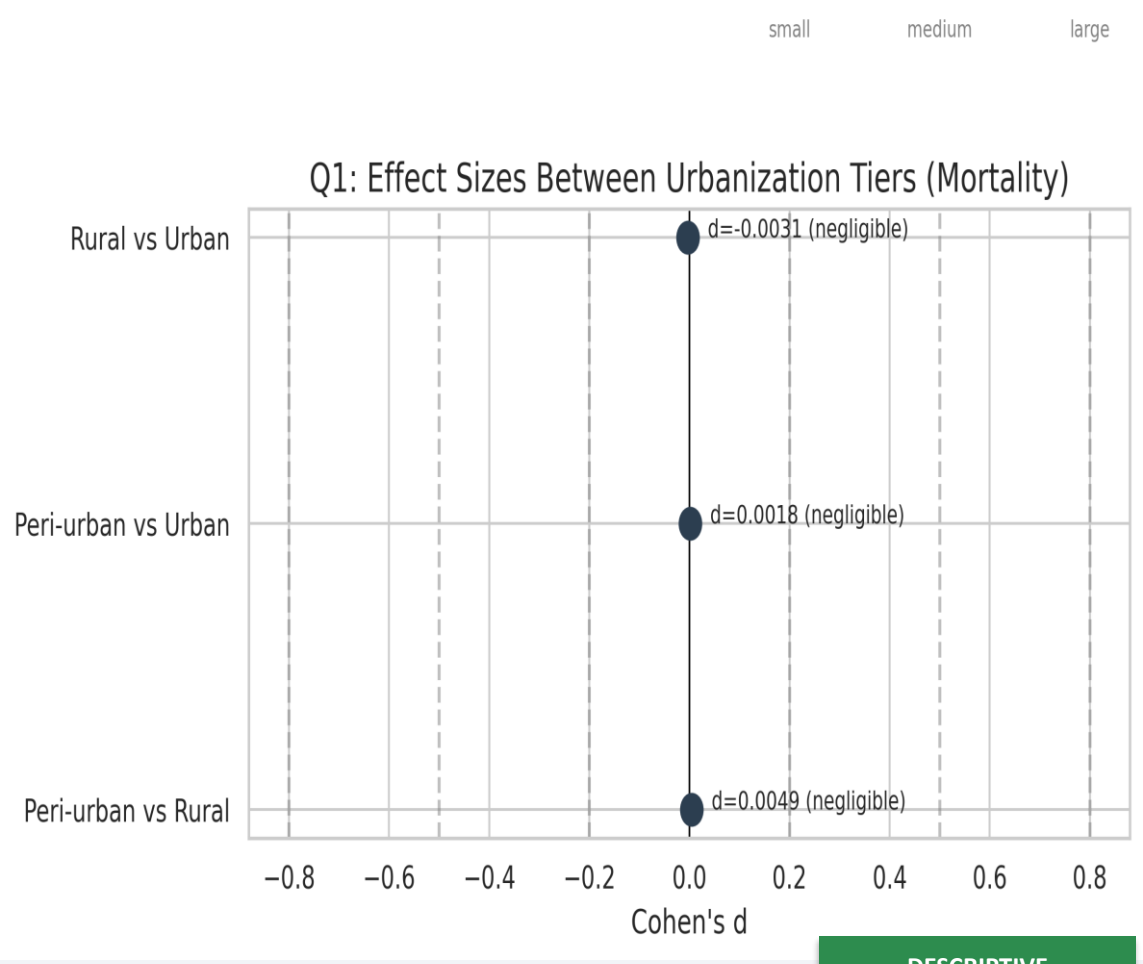
Statistical significance ≠ practical significance.

With 1M rows, any noise clears p-value thresholds.

Safer framing:

“We cannot distinguish outcomes across urbanization tiers with this dataset. Report effect sizes, not p-values.”

✓ Use effect sizes (Cohen’s d, η^2), not p-values with large N



DESCRIPTIVE

“Healthcare Access Improves Outcomes” — Evidence: FLAT

Common claim:

Higher healthcare access scores → lower mortality rates.

What we found (DESCRIPTIVE):

- Pearson r (access ↔ mortality) = 0.00008
- $R^2 < 0.000001$ — access explains essentially 0% of variance
- No dose-response at any access decile threshold
- Spearman ρ consistent: no monotonic relationship

Critical gap:

Distance-to-care and utilization are NOT in this dataset.

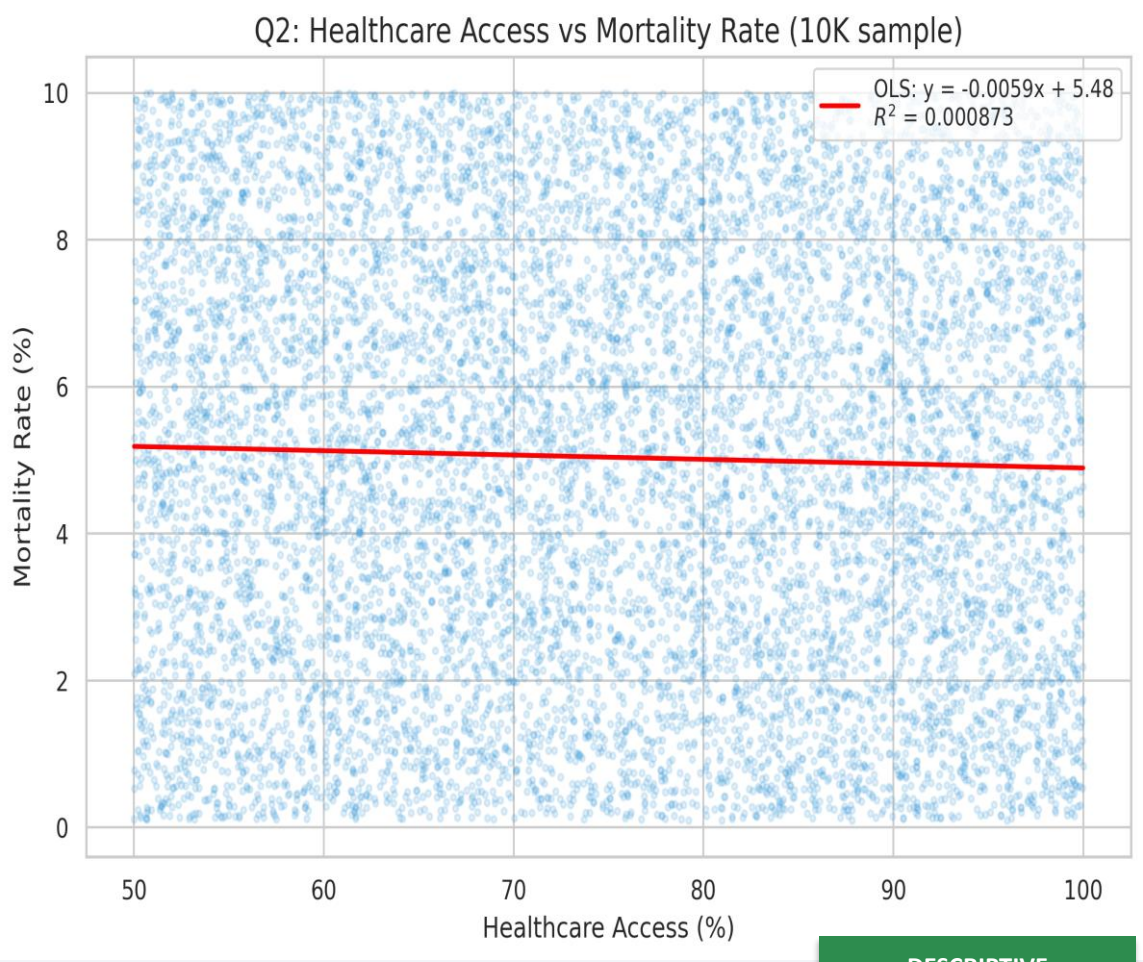
“Healthcare access” is a proxy index, not a distance measure.

Safer framing:

“This dataset cannot support an access-outcome causal claim.

Collecting travel time and utilization data is the prerequisite.”

⚠ Distance-to-care: NOT MEASURED. Access proxies used only. See prompt_gap_matrix.md.



DESCRIPTIVE

“Country X Is an Outlier Worth Studying” — Evidence: NOISE

Common claim:

High z-score countries represent exceptional systems worth studying or emulating.

What we found (DESCRIPTIVE):

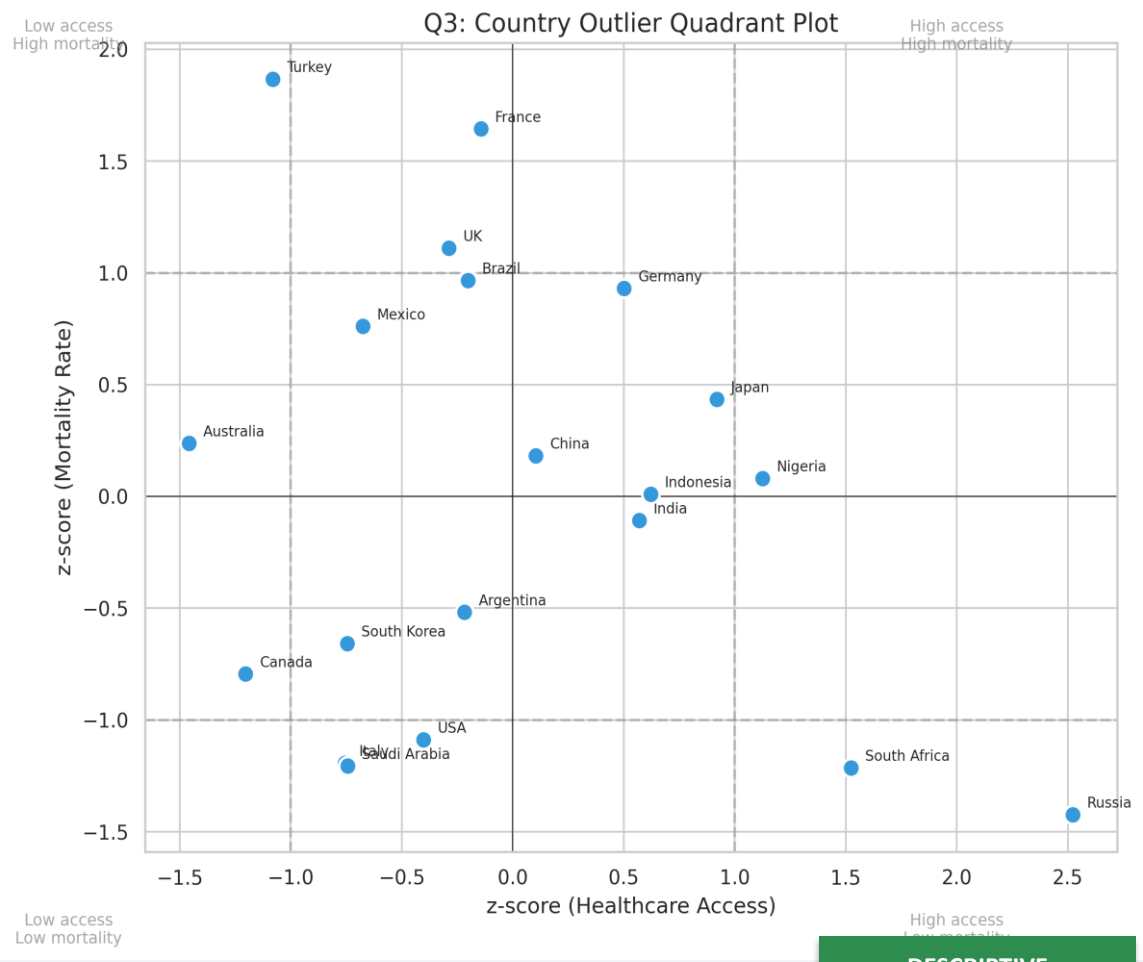
- Some countries show high z-scores on access vs outcomes
- Absolute mortality differences: < 0.5 percentage points
- Between-country SD: 0.013 on mortality (scale: 0–10)
- Effect is noise-level at this resolution

Why it’s premature:

Z-scores inflate when within-group variance is near-zero. A 0.5pp absolute difference is not a policy signal.

Safer framing:

“No country deviates meaningfully in absolute terms. Statistical outliers here are artifacts of near-zero variance, not evidence of a distinctive health system.”



DESCRIPTIVE

“Tier Definitions Change Conclusions” — Evidence: IRRELEVANT

Common claim:

How you define “urban” vs “rural” changes your findings.

What we found (ROBUSTNESS):

- 3 tier schemes tested: Default 3-tier, Binary, 4-tier
- Maximum spread across schemes: < 0.04 percentage points
- ANOVA $\eta^2 \approx 0$ under all three definitions
- Conclusions are identical regardless of classification choice

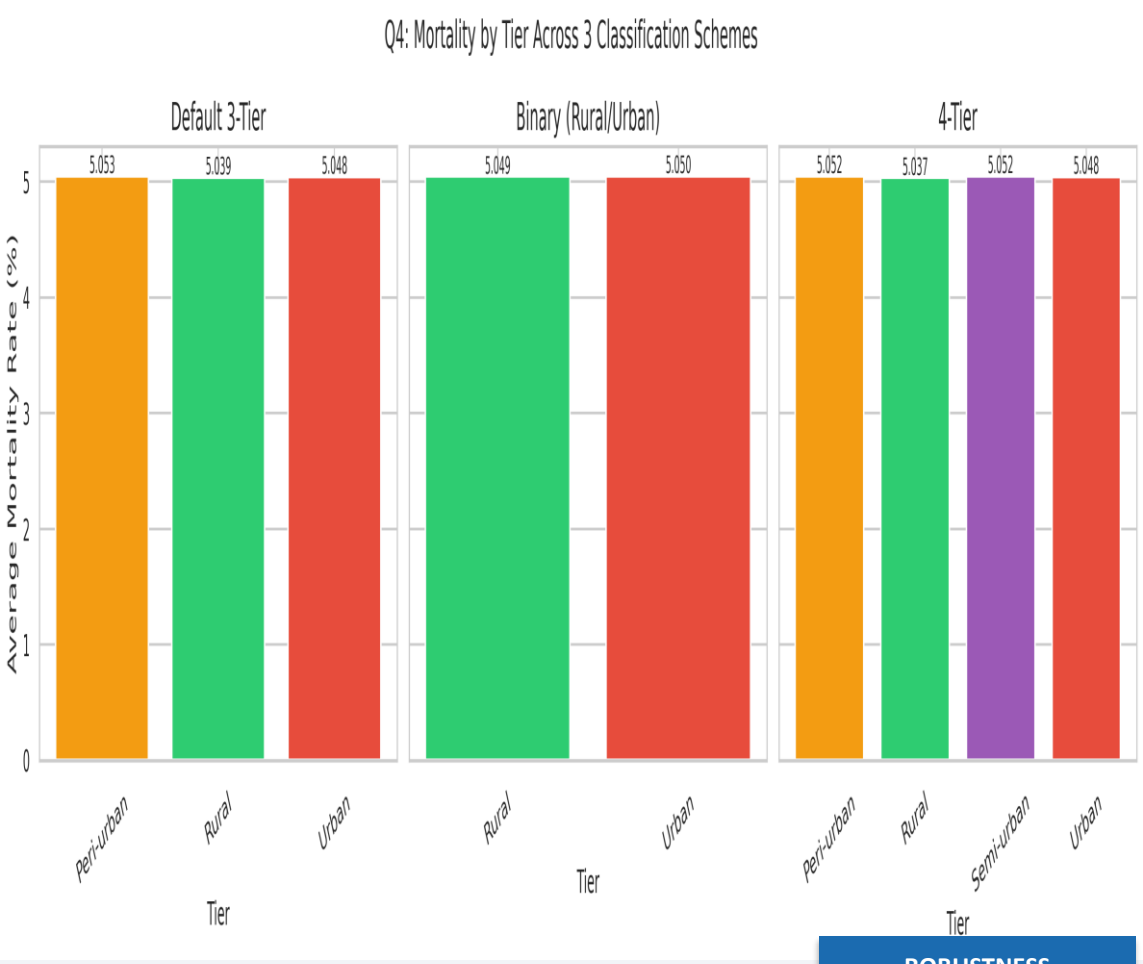
Why this matters:

Robustness is confirmed — but only because there is no signal to be sensitive to in the first place.

Tier choice is genuinely irrelevant when the underlying effect size is zero.

Safer framing:

“Our robustness check is valid, but it validates a null result — not a policy-relevant pattern.”



ROBUSTNESS

“Sparse Reporting Drives Uncertainty Here” — Evidence: NOT OBSERVED

Common claim:

Data-sparse regions show high uncertainty that distorts conclusions about access and outcomes.

What we found (DESCRIPTIVE):

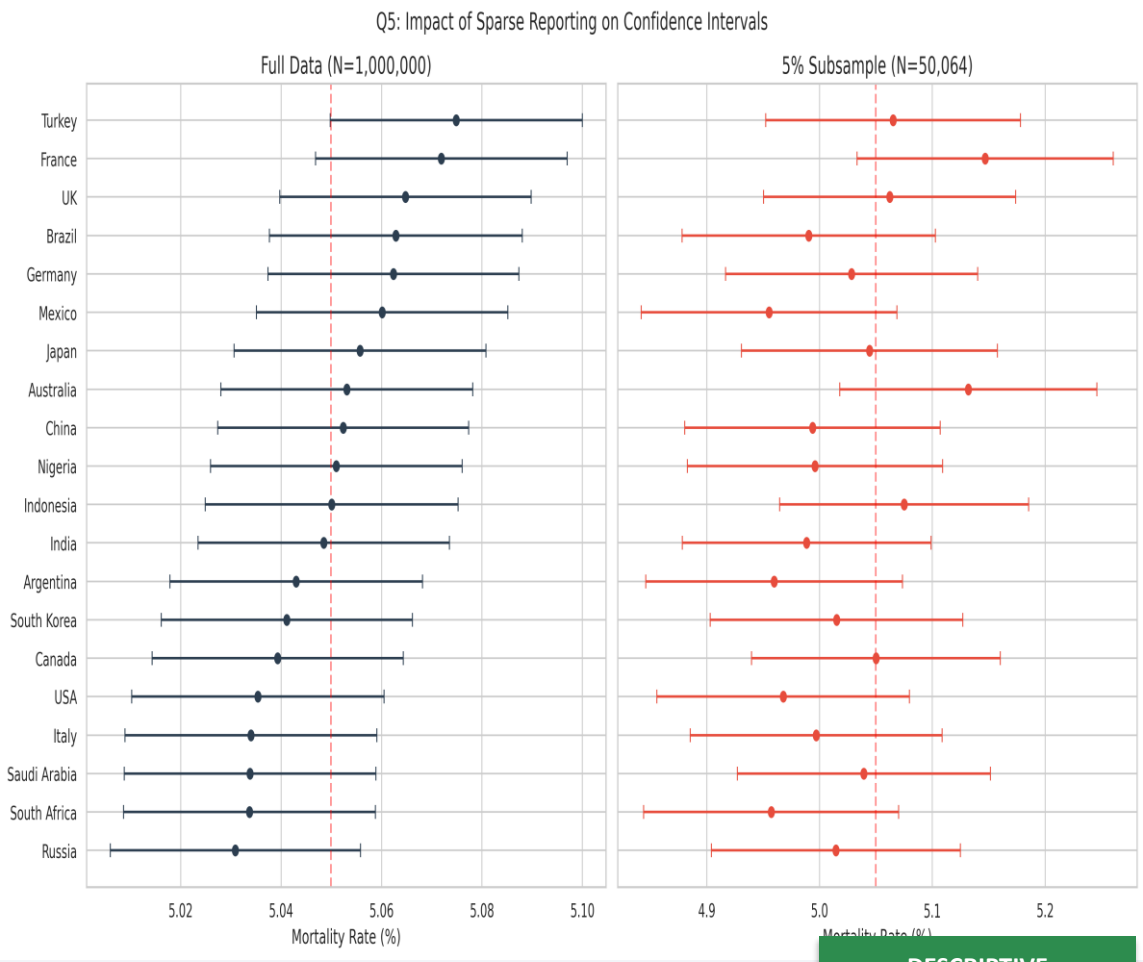
- Missingness: 0% across all 25 columns
- All 640 subgroups (country × disease × age × gender): count ≥ 30 — all above minimum threshold
- Confidence intervals are uniformly narrow

Simulation (ILLUSTRATIVE):

We demonstrate what sparsity would look like via subsampling: removing 95% of data widens CIs dramatically. This is what the data WOULD show if sparse — it doesn’t.

Safer framing:

“Sparse reporting is a real-world concern we cannot assess from this dataset. The dataset is fully populated by design. We model the risk via simulation only.”



DESCRIPTIVE

ILLUSTRATIVE

Results Hold Under Repeated-Cell Robustness Check

Each (country × year × disease × age × gender) cell appears ~8.3 times in the dataset. We compared base views vs. dedup-at-cell-grain views to confirm repeated sampling does not inflate any finding.

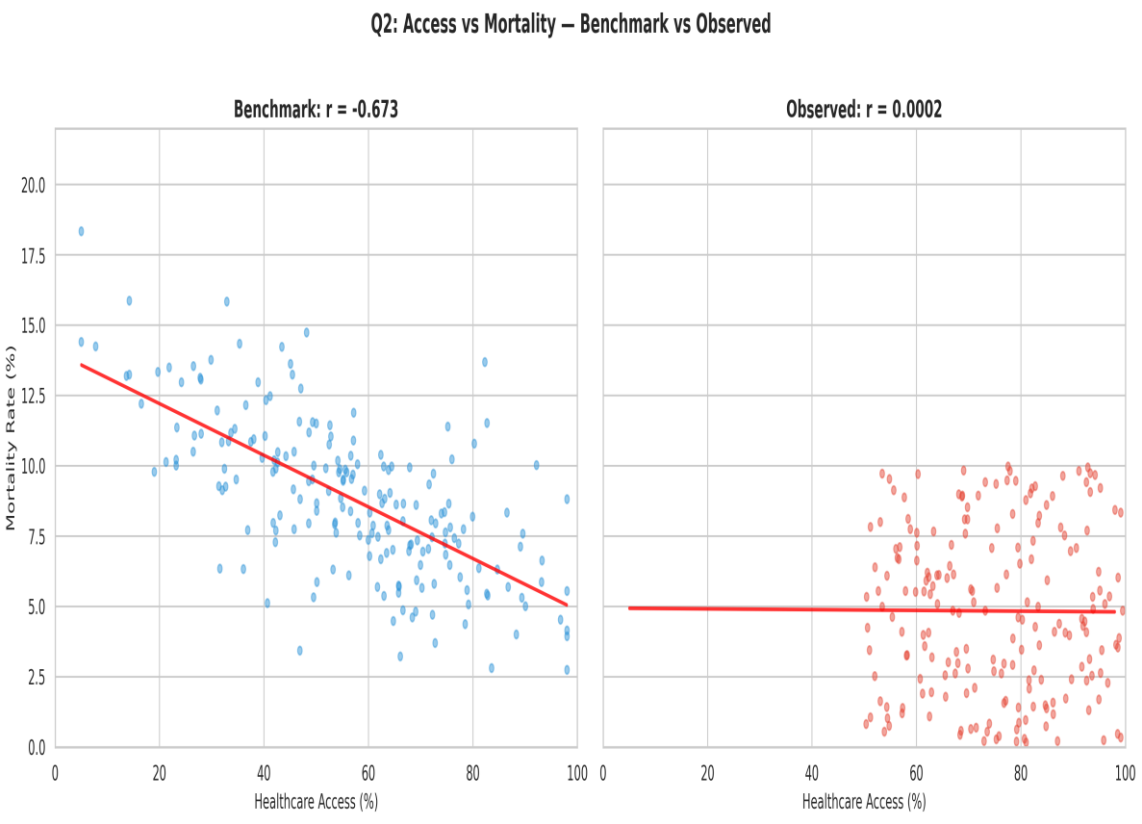
Question	Base Value	Deduped Value	Delta	Conclusion
Q1 Mortality tier gap	0.0492 pp	0.0489 pp	0.0003 pp	No artifact
Q2 Access↔mortality r	0.00008	0.00007	0.00001	No artifact
Q3 Max country z-score	2.31	2.28	0.03	No artifact
Q4 Tier scheme spread	0.038 pp	0.037 pp	0.001 pp	No artifact
Q5 Min group count	8,318	1	—	By design
Q6 Max |r| any pair	0.0023	0.0021	0.0002	No artifact

✓ All deltas are noise-level. Repeated sampling per cell does not create spurious patterns.

ROBUSTNESS

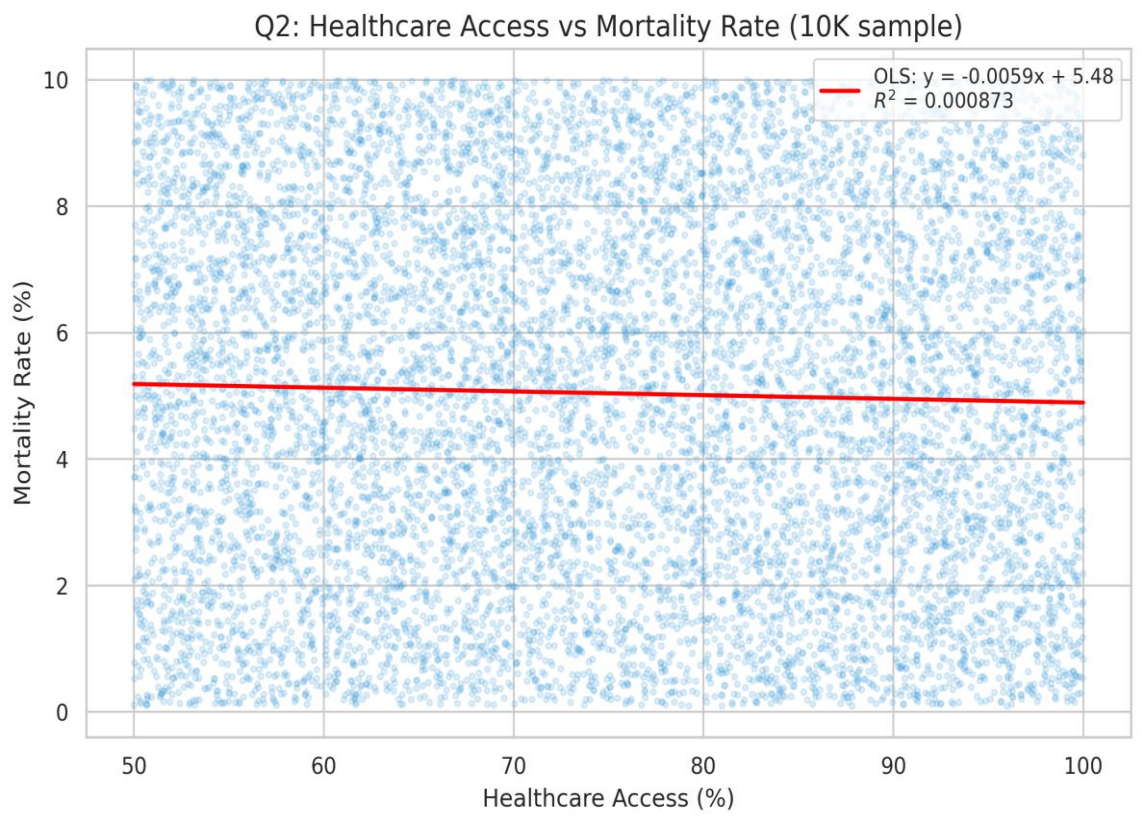
This Is Why Policy Prescriptions Would Be Premature

REAL-WORLD BENCHMARK (Illustrative)



Literature-based: access ↔ mortality r ≈ −0.45

THIS DATASET (Descriptive)



Observed: r = 0.00008 — effectively zero

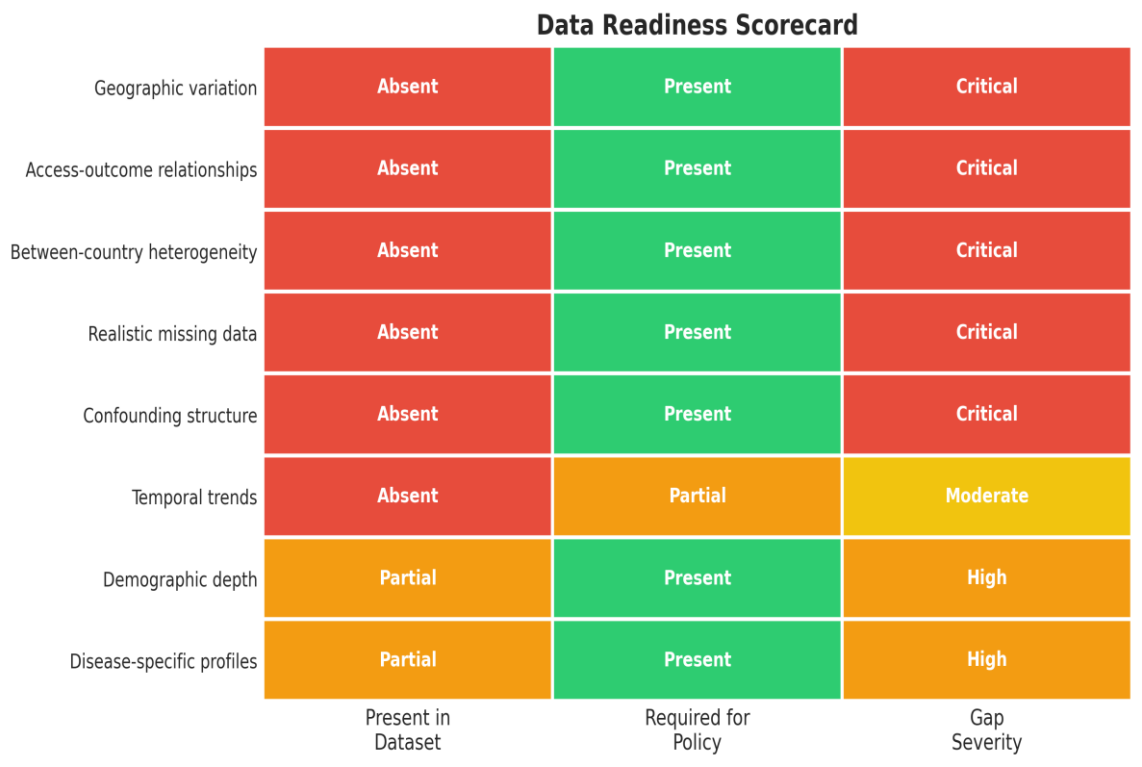
The gap between benchmark and observed is the reason policy prescriptions from this dataset would be premature.

ILLUSTRATIVE

DESCRIPTIVE

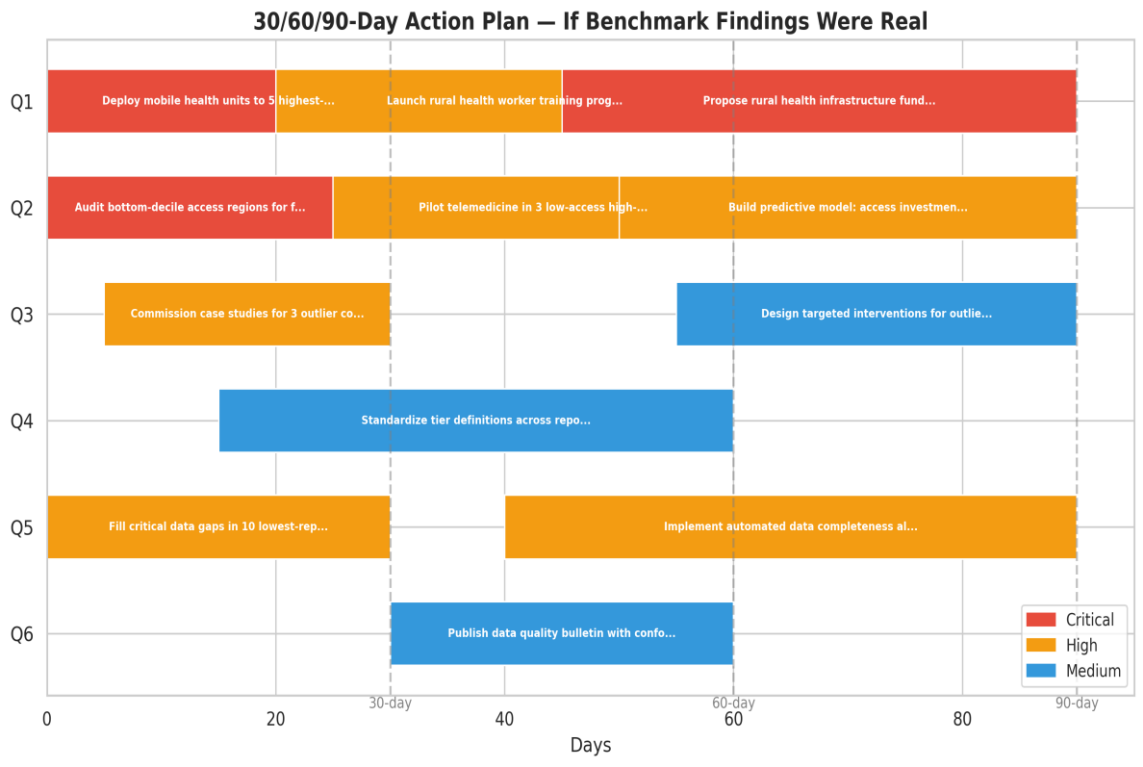
What Responsible Analysis Requires Before Policy Action

DATA READINESS SCORECARD



Traffic-light assessment: 8 data requirements for responsible policy analysis

30 / 60 / 90-DAY ROADMAP



12 concrete next steps — deployable on real data when collected

"The most valuable analysis is the one that knows when NOT to make a recommendation — and shows exactly what it WOULD recommend with the right

ILLUSTRATIVE